

# Deva Anand

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## Education

### University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

**Coursework:** Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

### Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

## Skills

### Languages

Python, C++, JavaScript, TypeScript, Java, SQL, Bash

### ML & Deep Learning

PyTorch, training workflows, optimizers & convergence, autodiff (from scratch), generative models, transfer learning, HuggingFace Transformers, NumPy, scikit-learn

### KG, RAG & Retrieval

Knowledge graphs, RAG, multi-view retrieval, RRF, cross-encoder reranking, MMR, sentence-transformers, multi-hop QA

### Evaluation & Reasoning

Metric design, discriminative-power objectives, dataset curation, statistical analysis (bootstrap CIs, McNemar), interpretability

### Performance & Data

Profiling (Spark UI), throughput / latency tuning, distributed-data processing, Apache Spark / PySpark, Docker, AWS / GCP / Azure

## Experience

### Wysa

July 2023 – July 2025

Backend Engineer

Bengaluru, India

- Architected full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by AI team to label data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

### Cario Growth Services Pvt. Ltd.

November 2022 – May 2023

Backend Developer Intern

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, and integrated LLM-powered context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience.

## Projects

### KG2RAG-Enhanced - Knowledge Graph + RAG

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama, sentence-transformers, PyTorch | UMass CS 685 Advanced NLP (team)

- Led the **multi-view retrieval** component of a team **knowledge-graph-guided RAG** pipeline for multi-hop QA (sub-question decomposition, per-view retrieval, RRF fusion + cross-encoder reranking + MMR), lifting supporting-fact recall **+2.68%** over the KG<sup>2</sup>RAG baseline on a **4,905-question** evaluation.
- Ran experiments over knowledge graphs of **28,492 entities** with token/latency logging and 95% binomial confidence intervals on an Ollama + LLaMA-3 pipeline.

### AutoEval - Reasoning Evaluation & Metric Design

github.com/Deva-1903/AutoEval

Python, evaluation methodology, Git automation

- Built an agent that performs **dataset curation and metric design**: it proposes test cases and labels, scores systems against a **discriminative-power** objective, and accepts only updates that improve it, with anti-Goodharting guardrails for verifiable, reproducible evaluation.

### Refusal Decay - LLM Interpretability & Evaluation

UMass CS 602 Research Project (graded full marks)

PyTorch, HuggingFace Transformers, Llama-3.1-8B-Instruct

- Ran a mechanistic-interpretability study of refusal in **Llama-3.1-8B** with causal interventions, multi-seed controls, and rigorous **metric design**, reporting a clean null with bootstrap 95% CIs and McNemar's test.

### Deep Learning Models & Optimization (CS 689)

github.com/Deva-1903/UMASS\_CICS\_689

PyTorch, JAX, NumPy | UMass CS 689 Advanced ML

- Trained **5 architectures across 3 optimizers** on CIFAR-10 with learning-rate tuning, implemented a reverse-mode **autodiff engine from scratch** (validated against JAX), and derived GD/SGD convergence guarantees.

## Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.